

Compiler Tomography

MDL identification of a shared compiler kernel from paired specification-realization data

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Date: August 3, 2026 **Status:** two theorems + one exact worked example. Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)) and *Concern as Fiber Geometry* ([papers/concern_as_fiber_geometry/paper.md](#)); depends on Theorem 1 of the parent paper (existence of the master fibration).

Abstract

The Structural Intelligence Conjecture (SIC) posits a stochastic fibration (q, K) with a *compiler* $K : Z \rightsquigarrow X$. In practice we rarely observe K directly; instead we see many pairs (s_i, x_i) where s_i is a specification and $x_i \sim K(\cdot \mid s_i)$ is a realization. This paper asks when — and how efficiently — the shared compiler K can be *identified* from such pairs, and gives two clean theorems answering the question in the finite discrete case.

- **Theorem CT-1 (MDL identification).** For a finite specification alphabet s , a finite realization alphabet x , and a family of compilers $\{K_\theta : \theta \in \Theta\}$ all sharing the marginal structure $K_\theta(\cdot \mid s) \in \Delta(q^{-1}(s))$, the Minimum Description Length (equivalently BIC) estimator over paired data $\{(s_i, x_i)\}$ is consistent: as the number of pairs grows, the estimated $\hat{\theta}$ concentrates on the true θ^* with probability one, at rate $O(\sqrt{(\log N) / N})$ in total variation.
- **Theorem CT-2 (Ecology dynamics: compiler improvement).** Under a reward-driven kernel update $K_{t+1}(\cdot \mid s) \propto K_t(\cdot \mid s) \cdot \exp(\beta \cdot r(s, x))$ with reward r bounded and $\beta > 0$ fixed, the expected reward under K_t is non-decreasing in t for every s (a discrete-time analogue of natural-gradient ascent on the reward functional). Fixed points of the update are exactly the reward-maximising kernels within the concern-parameterised family of the companion paper.

CT-1 provides the identifiability guarantee behind extended-program §5.6 ("compiler tomography") of the parent paper; CT-2 provides its dynamical counterpart ("compiler ecology"). Both are proved in the discrete case with no measure-theoretic machinery beyond finite Bayes rules and Wald- style consistency for MDL.

An exact instrument ([experiments/compiler_tomography_pair](#)) exhibits both theorems on the 4-bit Boolean world of Instrument 4.

1. Setup

Fix a finite specification alphabet s , a finite realization alphabet x , and a quotient map $q : X \rightarrow s$ (equivalently: $s = \text{image}(q)$ is the coarse structure the compiler renders). A **compiler kernel** is a Markov kernel $K : s \rightsquigarrow X$ with $\text{supp } K(\cdot \mid s) \subseteq q^{-1}(s)$ for every s .

Paired data. We observe N i.i.d. pairs $\{(s_i, x_i)\}_{i=1..N}$ where each $s_i \in s$ is drawn from an arbitrary specification distribution $P_S(s)$ and each $x_i \sim K^*(\cdot \mid s_i)$ from the *true* compiler K^* .

Family of hypotheses. We restrict attention to a finite (or countable) family $\mathcal{K} = \{K_\theta : \theta \in \Theta\}$ of candidate compilers. Each K_θ is parameterised by a finite string θ with description length $L(\theta)$ (bits under a chosen prefix-free code); MDL selects

$$\hat{\theta} = \operatorname{argmin}_{\{\theta \in \Theta\}} \{ L(\theta) + L(\text{data} \mid K_\theta) \},$$

where $L(\text{data} \mid K_\theta) = -\sum_i \log_2 K_\theta(x_i \mid s_i)$ is the code length of the data under K_θ .

2. Theorem CT-1: MDL identification (consistency)

Setup (CT-1). Suppose:

- The true compiler $K^* = K_{\{\theta^*\}}$ for some $\theta^* \in \Theta$.
- Θ is finite (or the description-length function $L(\theta)$ satisfies Kraft's inequality on a countable Θ).
- The specification distribution P_S has support covering S (i.e. every $s \in S$ is seen with positive probability).
- Identifiability: $\theta \neq \theta^* \Rightarrow K_\theta \neq K_{\{\theta^*\}}$ at some (s, x) with $P_S(s) > 0$ and $K_{\{\theta^*\}}(x | s) > 0$.

Theorem CT-1 (MDL consistency). Under the setup above, the MDL estimator $\hat{\theta}_N$ from N i.i.d. pairs satisfies

$$\Pr[\hat{\theta}_N = \theta^*] \rightarrow 1 \quad \text{as } N \rightarrow \infty,$$

and the excess code length satisfies

$$E[L(\hat{\theta}_N) + L(\text{data} | K_{\{\hat{\theta}_N\}}) - L(\theta^*) - L(\text{data} | K_{\{\theta^*\}})] = O(\log N).$$

In total variation, $dTV(K_{\{\hat{\theta}_N\}}, K_{\{\theta^*\}}) \rightarrow 0$ at rate $O(\sqrt{(\log N / N)})$ on the support of P_S .

Proof. Standard MDL / BIC consistency (Rissanen 1978, Barron–Cover 1991) specialised to the finite-conditional-alphabet case.

For any fixed $\theta \neq \theta^*$, by the identifiability assumption there exist s, x with $P_S(s) > 0$ and $K_{\{\theta^*\}}(x | s) \neq K_\theta(x | s)$. Then the KL divergence

$$\Delta_\theta := \sum_s P_S(s) \cdot KL(K_{\{\theta^*\}}(\cdot | s) || K_\theta(\cdot | s))$$

is strictly positive.

The empirical log-likelihood ratio between θ^* and θ is $-(L(\text{data} | K_\theta) - L(\text{data} | K_{\{\theta^*\}})) / \log 2 = \sum_i \log(K_{\{\theta^*\}}(x_i | s_i) / K_\theta(x_i | s_i))$. Its expectation is $N \cdot \Delta_\theta$, and by Hoeffding's inequality it exceeds $N \cdot \Delta_\theta / 2$ with probability at least $1 - \exp(-N \cdot \Delta_\theta^2 \cdot c)$ for a constant $c > 0$ depending on the finite alphabet.

MDL selects $\hat{\theta}$ iff $L(\hat{\theta}) + L(\text{data} | K_{\{\hat{\theta}\}}) \leq L(\theta^*) + L(\text{data} | K_{\{\theta^*\}})$. For $\theta \neq \theta^*$, this fails as soon as $L(\text{data} | K_{\{\theta^*\}}) - L(\text{data} | K_\theta) < L(\theta) - L(\theta^*)$, i.e. as soon as the empirical log-likelihood ratio exceeds $L(\theta) - L(\theta^*)$ (a constant). Combined with Hoeffding: $\Pr[\hat{\theta} = \theta] \leq \exp(-N \cdot \Delta_\theta^2 \cdot c / 2)$ for large enough N .

Union-bounding over the finite Θ (or, in the countable case, using Kraft's inequality to bound $\sum_\theta 2^{-L(\theta)}$):

$$\Pr[\hat{\theta}_N \neq \theta^*] \leq \sum_{\theta \neq \theta^*} \exp(-N \cdot \Delta_\theta^2 \cdot c / 2) \rightarrow 0.$$

Total variation rate follows from Le Cam's inequality $dTV(K_{\hat{\theta}}, K_{\{\theta^*\}})^2 \leq \frac{1}{2} \cdot KL(K_{\hat{\theta}} || K_{\{\theta^*\}})$ and the fact that the MDL score gap is $O(\log N)$ in expectation, so the surviving θ -candidates have $KL(K_\theta || K_{\{\theta^*\}}) = O(\log N / N)$. Taking the square root gives the $O(\sqrt{(\log N / N)})$ rate. $\square$

Consequence (operational). The compiler-tomography extended-program target of the parent paper (§5.6) is a theorem in the finite discrete case: MDL is a consistent estimator of the shared compiler from paired specification-realization data, at parametric rate.

The specialisation to Boolean fiber-preserving compilers on the 4-bit world of Instrument 4 is worked out in §4 of this paper and verified by the companion instrument (§5).

3. Theorem CT-2: Ecology dynamics (compiler improvement)

The extended-program §5.6 also names *compiler ecology* $K_{\{t+1\}} = U(K_t, \text{outcomes})$: a compiler that *updates* toward good outcomes. The natural update rule that plays this role is the concern-parameterised reweighting of the

companion paper *Concern as Fiber Geometry* Theorem CG-1, with the reward serving as concern:

$$K_{t+1}(dx \mid s) = K_t(dx \mid s) \cdot \exp(\beta \cdot r(s, x)) / \Xi_t(s),$$

where $r : S \times X \rightarrow [0, R]$ is a bounded reward, $\beta > 0$ is a fixed sharpness, and $\Xi_t(s) := \int_{q^{-1}(s)} \exp(\beta r(s, x)) K_t(dx \mid s)$ is the fiber partition function.

Theorem CT-2 (Monotone compiler improvement). *For every $s \in S$ and every $t \geq 0$,*

$$E_{K_{t+1}}[r(s, X) \mid s] \geq E_{K_t}[r(s, X) \mid s],$$

with equality iff $r(s, \cdot)$ is $K_t(\cdot \mid s)$ -a.s. constant on $q^{-1}(s)$. Equivalently, the reward expectation is non-decreasing under compiler ecology; fixed points are exactly the compilers concentrated on $\operatorname{argmax}_x r(s, x)$ within each fiber (up to reward ties).

Proof. Fix s . Let $r(x) := r(s, x)$ and $p_t(x) := K_t(x \mid s)$ on $q^{-1}(s)$. The update is $p_{t+1}(x) = p_t(x) \cdot \exp(\beta r(x)) / \Xi, \Xi = E_{p_t}[e^{\beta r}]$. Then

$$\begin{aligned} E_{p_{t+1}}[r] &= E_{p_t}[r] \\ &= \sum_x r(x) \cdot p_t(x) \cdot (e^{\beta r(x)} / \Xi - 1) \\ &= (1/\Xi) \cdot \sum_x r(x) \cdot p_t(x) \cdot (e^{\beta r(x)} - \Xi) \\ &= (1/\Xi) \cdot [\sum_x r(x) e^{\beta r(x)} p_t(x) - \Xi \cdot \sum_x r(x) p_t(x)] \\ &= (1/\Xi) \cdot [E_{p_t}[r \cdot e^{\beta r}] - E_{p_t}[e^{\beta r}] \cdot E_{p_t}[r]] \\ &= (1/\Xi) \cdot \operatorname{Cov}_{p_t}[r, e^{\beta r}]. \end{aligned}$$

$\operatorname{Cov}(r, e^{\beta r}) \geq 0$ always, because $r \mapsto e^{\beta r}$ is monotone increasing for $\beta > 0$ (positive-correlation inequality — a special case of the FKG / Chebyshev correlation inequality for a *single* random variable being compared with a monotone function of itself). Equality holds iff r is p_t -a.s. constant on the support (so that $\operatorname{Cov}(r, f(r)) = 0$ for any f).

Iterating: $E_{K_t}[r \mid s]$ is non-decreasing in t . Since it is bounded above by $\max_{x \in q^{-1}(s)} r(s, x) \leq R < \infty$, it converges to a limit. The limit is a fixed point iff $\operatorname{Cov}_{p_\infty}[r, e^{\beta r}] = 0$, i.e. iff $r(s, \cdot)$ is p_∞ -a.s. constant on $q^{-1}(s)$; since p_∞ is concentrated on $q^{-1}(s)$, this forces p_∞ to be supported on a single reward level within the fiber. The maximal such level is $\max_x r(s, x)$, achieved when p_∞ concentrates on argmax . $\square$

Consequence (operational). Under the Boltzmann-type reweight $K_{t+1} \propto K_t \cdot e^{\beta r}$, the compiler ecology of the parent paper's §5.6 is a strict monotone-improvement dynamics on reward, with argmax fixed points. For $\beta \rightarrow \infty$ this becomes pure exploitation (delta on argmax); for $\beta \rightarrow 0$ it becomes the identity update (no change per step). Intermediate β gives the standard exploration-exploitation trade-off familiar from tempered posteriors / soft-Q iterations.

4. Worked example: 4-bit Boolean world

World. $X = \{0, 1\}^4, q(x) = (x_0 \oplus x_1, x_2 \oplus x_3) \in S, |S| = 4$. Base compiler $K^*(\cdot \mid s)$ is uniform on the 4-element fiber $q^{-1}(s)$.

Hypothesis family for CT-1. We take $\theta = \{(\theta_1, \theta_2) \in \mathbb{R}^2\}$ with the concern parameterisation of *Concern as Fiber Geometry* — each θ corresponds to the reweighted compiler $K_\theta(x \mid s) \propto K^*(x \mid s) \cdot \exp(\theta_1 (2x_0 - 1) + \theta_2 (2x_2 - 1))$, i.e. a two-dimensional exponential family with sufficient statistic $T(x) = (2x_0 - 1, 2x_2 - 1) \in \{-1, 1\}^2$. The true parameter is $\theta^* = (0, 0)$ (uniform base compiler).

MDL identification on this family. For N paired samples, MDL on a finite grid $\theta = \{(\theta_1, \theta_2) : \theta_i \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}\}$ (25 candidates each with equal description length $\log_2 25$) selects $\hat{\theta}_N$. Because the true θ^* is inside the grid and every other grid point has a positive KL divergence to it, Theorem CT-1 predicts $\Pr[\hat{\theta}_N = \theta^*] \rightarrow 1$ as $N \rightarrow \infty$.

Reward for CT-2. Take $r(s, x) := x_0 + x_2$ (a bounded scalar reward in $\{0, 1, 2\}$). Starting from $K_0 = K^*$ (uniform on each fiber), the update $K_{t+1} \propto K_t \cdot e^{\beta r}$ with $\beta = 1$ should monotonically increase $E[r \mid s]$ for each s until it converges to the argmax reward on the fiber.

The companion instrument computes both trajectories exactly.

5. Instrument: `experiments/compiler_tomography_pair`

Exact witness of Theorems CT-1 and CT-2 on the setup above:

- **CT-1.** For $N \in \{50, 100, 200, 500, 1000, 2000\}$ paired samples, run MDL over the 25-point grid. Report the empirical $\Pr[\hat{\theta}_N = \theta^*]$ averaged over 100 seeded trials, and verify it exceeds $1 - 0.05$ for $N \geq N_{\text{threshold}}$ (pre-registered, determined by the finite-KL-gap calculation).
- **CT-2.** Iterate the compiler ecology update for $T = 20$ steps at three β values $\{0.1, 1.0, 4.0\}$ starting from $K_0 = K^*$, and verify $E_{\{K_t\}}[r \mid s]$ is monotone non-decreasing in t at every (s, β) and converges to the fiber argmax at $\beta = 4$.

Both experiments are deterministic under fixed seed, and gate pass/fail is exact (either monotone or not; either recovery-rate ≥ 0.95 or not).

6. Relation to the SIC framework

Theorem CT-1 promotes the extended-program §5.6 target (compiler tomography) from a *research direction* to a *theorem* in the finite discrete case, via classical MDL consistency. Theorem CT-2 does the same for the "compiler ecology" clause of §5.6.

Together with Theorem 7 (linear-ICA and now sparse-ICA / IMA positive resolutions of SIC-C-c, Instruments 8 and 9 of the parent paper) and Theorems CG-1, CG-2 (Fisher geometry and holonomy of concern from the companion paper), the SIC extended program now has *five* explicit theorem-instrument pairs beyond the six of the parent paper. The remaining §5 constructs (theory atlas §5.5, causal semantics §5.7, representation-repair calculus §5.8, alignment as ensemble governance §5.9, autocatalytic artwork §5.10) remain open.

7. Limitations

- The MDL consistency theorem CT-1 requires *finite alphabets* on both sides (s and x) and a countable hypothesis family Θ with a well-defined description-length code. The extension to continuous x requires a smoothness / prior-mass argument (Barron 1998); not proved here.
 - The compiler ecology theorem CT-2 requires a bounded reward and a fixed β . Time-varying β_t (annealing) or unbounded rewards need additional convergence arguments; classical convergence-rate results for tempered-Bayes / soft-Q are the appropriate literature.
 - Neither theorem is Lean-formalised yet. The union-bound machinery from Theorem 5's Lean proof is not enough; a mathlib-backed formalisation of BIC / Wald consistency would be needed for CT-1.
 - The identifiability assumption for CT-1 rules out the observational equivalence class of compilers — two compilers agreeing on every (s, x) pair are considered identical. Real-world compilers often have such equivalence classes (e.g. by symmetry); the theorem identifies up to that equivalence.
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8. Reproduction

```
python3 experiments/compiler_tomography_pair/experiment.py
```

Full development is in the parent paper's §5.6 and the notes file `notes/structural_intelligence_conjecture.md`.